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Standardization of Gas Lift Manifold for the Enhancement of Design Related to Maintenance, Operability, and Accessibility

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Abstract

Offshore Industry has always been very challenging since the inception of it. Even after many decades of Offshore Platform design, COMPANY has been trying to standardize its Well Head Platform Design to ensure that it is a step ahead from the normal Well Head Platform conventional design.

The reason for the proposed standardization is due to the current Piping configuration, which has been observed to contribute to flow inefficiencies, increased pressure losses, and restricted access to key operational valves, which further complicates routine maintenance and poses risks during emergency situations. The proposed redesign standardization introduces an optimized layout that prioritizes improved configuration that not only supports smoother gas flow and system reliability but also enables faster response times during shutdowns and startups. It significantly reduces downtime during maintenance activities and enhances the overall maintainability of the system. The design also provides improved spatial efficiency, enabling safer movement for personnel and better integration with surrounding equipment in line with industry standards such as ASME B 31.3 and API RP 14E.

Overall, the implementation of this new layout arrangement is expected to deliver measurable benefits in terms of Project execution time, reduced operating costs, increased uptime, improved system integrity, and compliance with regulatory and safety requirements providing long-time benefits during maintenance. It also presents a forward-looking solution that supports the long-term performance and sustainability of gas lift processing operations.

Introduction

The offshore oil and gas industry operates in one of the most technically challenging and risk-intensive environments in the world. Also, this industry has always presented formidable engineering and operational challenges. Despite decades of innovation and development in offshore platform design, continuous improvements remain essential to meet evolving demands in safety, efficiency, and cost optimization.

COMPANY has achieved considerable success in standardizing its wellhead platform designs for both oil production and water injection operations. This ongoing standardization reflects COMPANY's commitment

to innovation and operational excellence, positioning its Gas lift platforms ahead of conventional offshore infrastructure.

Extracting hydrocarbons from beneath the seabed requires not only robust and reliable equipment but also optimized design strategies that ensure maximum productivity, minimal downtime, and uncompromised safety. Among these strategies, standardization of equipment and system design plays a pivotal role in enhancing the efficiency and effectiveness of offshore operations.

This paper discusses the standardization of the gas lift manifold system as part of a broader effort to improve the design of offshore wellhead platforms, particularly in relation to maintenance, operability, and accessibility. The proposal addresses the new optimized layout configuration in gas lift design and aims to streamline operations, enhance maintenance, accessibility and reduce risks, and align with best industry practices.

Overview

To sustain and enhance production levels while optimizing the production process in terms of efficiency and environmental compliance, various system expansions are periodically implemented. One such initiative is the integration of gas lift systems within existing fields which remains a critical method for enhancing oil recovery, particularly in mature reservoirs.

Gas lift technology is a widely adopted artificial lift technique in offshore production facilities. It involves the injection of high-pressure gas into the production tubing, which reduces the hydrostatic pressure of the produced fluids and thereby facilitates the upward flow of oil. This method is especially valuable in mature fields where natural reservoir pressure is no longer sufficient to push hydrocarbons to the surface.

Despite its widespread application, gas lift systems often suffer from design inefficiencies, particularly when integrated into aging infrastructure or rapidly developed fields without adequate planning for future expansions. In many offshore platforms, gas lift manifolds have evolved organically over time, leading to non-standardized layouts that complicate routine operations and maintenance. Also, the existing piping configurations in many offshore facilities exhibit inefficiencies that hamper operational performance. Such limitations not only impede routine maintenance but also compromise safety during emergency shutdowns. This necessitates a standardized approach to ensure safe and efficient integration that adheres to best practices and regulatory standards.

COMPANY has made significant progress in standardizing various aspects of its offshore platforms, including wellhead and water injection platforms. By doing so, the organization not only reduces capital and operational expenditures (CapEx and OpEx) but also ensures that its facilities adhere to the latest safety and performance standards. Building on that foundation, the standardization of the gas lift manifold presents an opportunity to further improve system reliability, personnel safety, and lifecycle costs.

Methodology

The key motivation or methodology behind standardizing the gas lift manifold design stems from the numerous operational inefficiencies and maintenance difficulties currently faced and hampering Operations. These include:

- Complex and inefficient piping layouts that result in high pressure drop and imbalances.
- Non accessibility to critical valves and components, posing risks during emergency operations and routine servicing.
- Excessive use of fittings and elbows, leading to turbulence and flow instability.
- Inadequate slope gradients in pipelines, that contribute to accumulation of liquid pooling, and increased risk of internal corrosion.

- Lack of clear separation between high-pressure and low-pressure lines, raising operational safety concerns.

In addressing these issues, the proposed standardization aims to deliver a manifold system that is both technically superior and operationally practical, leveraging best practices in pipeline design and valve placement to support long-term platform integrity. The manifold system, though often underestimated, lies at the heart of production assurance strategies in artificial lift systems.

This is not just a technical adjustment but a strategic decision that impacts project execution, safety culture, operational flexibility, and cost-efficiency. It aims to ensure design uniformity, predictability in operations, and ease of replicability across new developments. It enhances the ability to benchmark performance, track failures, and implement systemic improvements.

Design Objectives

The primary objectives of the proposed gas lift manifold standardization are:

1. Optimize flow efficiency by minimizing pressure drops and turbulence.
2. Enhance accessibility to operational components such as isolation valves, blowdown valves, and drain valves.
3. Reduce maintenance downtime through improved layout and clear valve segregation.
4. Ensure compliance with industry codes such as ASME B31.3, API RP 14E, and other applicable standards and recommended practices.
5. Improve personnel safety through better equipment layout and access pathways.
6. Facilitate future scalability by incorporating flexibility in the design for future tie-ins or expansions.

These objectives are defined through lessons learned in previous field operations, incident investigations, and performance audits. They reflect a proactive approach to mitigating common failure modes and aligning with emerging industry benchmarks.

Key Design Features.

Optimized Piping Layout

The new manifold design employs direct routing principles to eliminate unnecessary bends and elbows. By reducing the number of fittings, pressure losses are minimized, and smoother gas flow is achieved. The use of larger radius bends and straight runs enhances flow characteristics and minimizes erosion risks.

Strategic Valve Placement

Isolation, blowdown, and drain valves are positioned based on their operational importance and frequency of use. Valves that require regular manual operation are placed at waist height and within arm's reach from access platforms or walkways. Color coding and tagging systems are used to distinguish between high-pressure and low-pressure systems, enhancing situational awareness and reducing the potential for operator error.

Slope Control and Drainage

All gas lift lines are laid out with proper slope gradients to facilitate natural drainage of condensates. This helps prevent liquid accumulation and internal corrosion in gas systems. Additionally, slope orientation supports effective blowdown procedures during maintenance or shutdown events.

Space Utilization and Layout

The redesigned manifold configuration allows for better spatial efficiency, making room for safe personnel movement and easier integration with nearby systems. Equipment is placed considering clearances for lifting and removal, as well as minimizing obstruction to emergency egress routes.

Segregation of Pressure Zones

The design clearly separates high-pressure gas lift lines from low-pressure return lines and utility piping. This separation is both physical and functional, ensuring that maintenance work on one system does not

inadvertently impact the operation of another. Such segregation is essential for permit-to-work systems and improves operational risk management.

Operational and Maintenance Benefits.

Improved Safety

The enhanced layout reduces exposure to high-pressure equipment during maintenance and operation. The clear valve arrangement and labeling reduce the likelihood of human error. Additionally, easier access minimizes the time personnel spend in hazardous areas, thus limiting their exposure to potential process upsets and mechanical hazards. The segregation of zones further ensures that accidental operation or exposure to high-pressure systems is minimized.

Reduced Downtime

By simplifying valve access and improving piping layout, the need for full system shutdowns during routine maintenance is minimized. Blowdown and isolation valves enable sectional shutdowns, which keeps the rest of the system operational and reduces production interruptions. This partial shutdown not only improves availability but also contributes to enhanced production efficiency over time.

Lower Maintenance Costs

Elimination of redundant fittings and improved drainage minimizes corrosion, reducing the frequency and cost of repairs. The standardized parts and layout also allow for faster sourcing and replacement during maintenance activities. Bulk procurement of standardized components further reduces procurement cycles and ensures ready availability of spares.

Enhanced Reliability and Uptime

Smooth gas flow, optimized valve placement, and slope control contribute to fewer unplanned shutdowns and higher system availability. The standardization ensures consistent performance across platforms, which is crucial for centralized monitoring and control systems. Greater system reliability ultimately contributes to enhanced recovery rates, reduced flaring events, and improved environmental performance.

Predictive Maintenance Integration

Standardization allows for easier integration of sensor technology and predictive analytics, enabling a shift from reactive to proactive maintenance practices. For instance, vibration sensors on valves or flow meters can detect early anomalies, prompting timely intervention before failure occurs.

Simplified Training and Familiarization

Standardizing the manifold layout means that operators and maintenance personnel can be trained on a single design, reducing the learning curve and potential for error when moving between platforms or assignments. Training programs can be streamlined across assets, and skill competency frameworks can be unified. This harmonization results in cost savings, enhanced operational readiness, and lower incident rates.

Kindly refer to [Fig.1](#); which shows the Overall Operational Outlook on the Benefits of Standardization, which includes the parameters related to Improved Safety, Reduced Downtime, Lower Maintenance Cost and Improved Quality With and Without Standardization. These parameters have been derived with respect to In-house Operational field data.

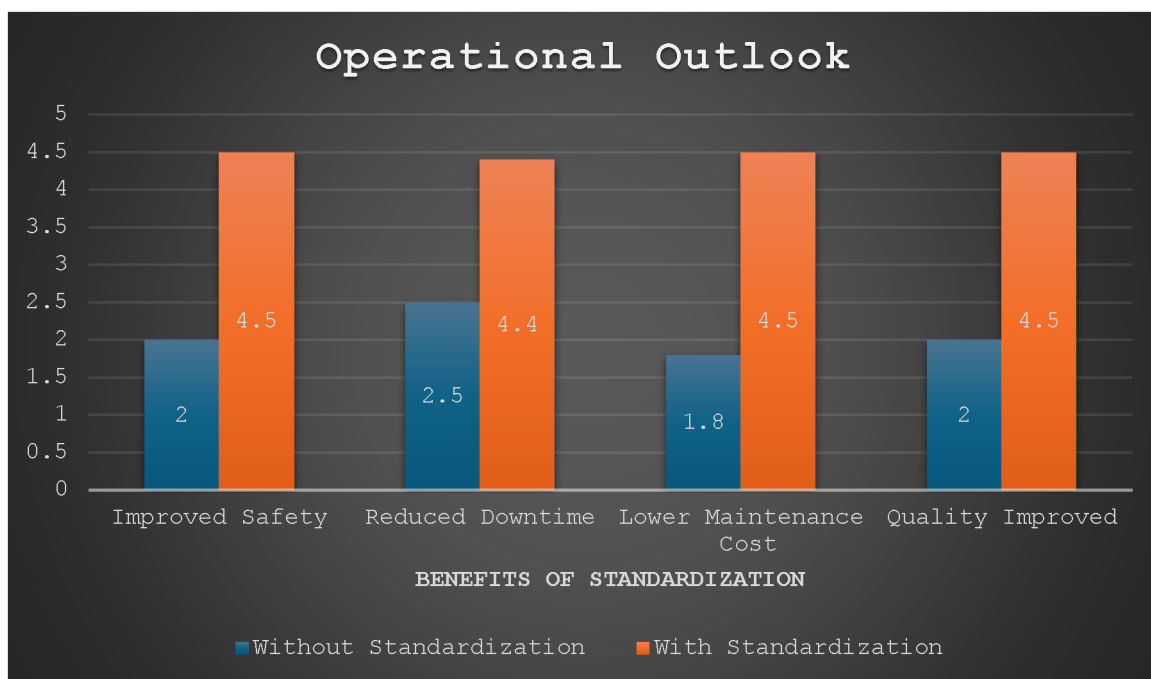


Figure 1—Overall Operational Outlook on the Benefits of Standardization.

Implementation Considerations. Implementing the standardized gas lift manifold involves several key activities that span engineering, procurement, construction, commissioning, and operations. These activities provide significant operational advantages and require close collaboration across multidisciplinary teams as mentioned:

- Site Survey and Assessment – Evaluate current manifold configurations, platform layouts, and access limitations. The assessment phase should include 3D scanning of existing piping systems, condition-based monitoring, and risk assessments to determine legacy system constraints.
- Design Adaptation – Customize the standardized design to fit platform-specific constraints while maintaining core features. Engineering teams must consider the brownfield conditions, existing tie-in points, and available footprint when adapting the base model. Flexibility should be retained for future debottlenecking.
- Stakeholder Engagement – Involve operations, maintenance, safety, and engineering teams to validate and refine the proposed layout. Lessons learned from past incidents should be incorporated through HAZOP and Constructability Reviews. Early engagement helps build ownership and minimizes change order requests later.
- Fabrication and Installation – Use modular fabrication where possible to reduce offshore construction time. Prefabricated spools and skids improve quality control and reduce hot work on the platform. Construction work packs should be detailed to reduce ambiguity.
- Testing and Commissioning – Ensure the new layout meets pressure test, blowdown, and leak detection requirements. Integrated FAT and SAT procedures should be conducted, with simulated operational scenarios to validate performance.
- Documentation and Training – Update system documentation including P&IDs, isometrics, ITPs, and redline markups. Personnel should be trained via interactive modules, and on-site familiarization sessions.

Successful implementation not only brings immediate gains in reliability and safety but also reinforces a continuous improvement culture. A well-documented rollout can be used as a reference case for other COMPANY assets.

Future Outlook and Scalability. Standardization is not a one-time initiative but a cornerstone of COMPANY's long-term asset management and digital transformation strategies. As offshore assets evolve, so too must the systems that support them.

With the global push toward digitalization, the standardized gas lift manifold can serve as a foundation for the following developments:

- Smart Instrumentation Integration – Integration of smart transmitters, solenoid-actuated valves, and diagnostics-enabled controllers that provide real-time feedback on system health and performance.
- Digital Twin Deployment – Digital Twin can be used for real time monitoring of the Gas Lift operations, providing intelligent insights and educating the operators by creating a digital replica of the manifold system that can simulate performance, track degradation, and support operator training in a risk-free environment. By simulating the manifold design, digital twin helps in predicting the fail scenarios which further reduces unplanned operational downtime and also supports scenario planning and asset lifecycle extension.
- Predictive Maintenance and AI – To address the limitations, the study advocates for hybrid models that integrate multiple AI techniques or combine with AI with numerical methods using machine learning algorithms to detect anomalies in flow patterns, valve behaviors, or pressure readings. These insights can trigger alerts, optimize maintenance schedules, and avoid unscheduled downtime enhancing efficiency and reliability, offering a more comprehensive solution and long-term economic viability.
- Automation and Remote Operations – Enhancing remote operations by integrating manifolds with SCADA systems, enabling remote isolation, drain, or blowdown without personnel intervention. This reduces offshore exposure and enhances resilience during emergencies.
- Replicability and Design Rollout – Once proven, the standardized design can be rolled out across brownfield upgrades, greenfield developments, or even new business ventures. This reduces engineering time and accelerates the first oil.

Standardization is a continuous improvement strategy, and the future of offshore production lies in modular, intelligent, and adaptable systems. By initiating this standardization today, COMPANY is setting the stage for operational resilience, cost competitiveness, and environmental stewardship tomorrow.

A long-term vision may include incorporating predictive maintenance tools, where sensor data from valves and flow meters can be analyzed to anticipate failures or performance degradation before they occur. The simplified and consistent design makes it easier to implement such digital solutions.

Environmental and Sustainability Considerations

In the modern offshore oil and gas landscape, environmental stewardship is no longer optional—it is a core operational imperative. The standardization of the gas lift manifold directly supports COMPANY's environmental and sustainability goals by enabling cleaner, more efficient, and more reliable operations.

Reduced Emissions and Flaring

One of the major contributors to greenhouse gas (GHG) emissions in offshore production is unplanned flaring during equipment malfunction or maintenance shutdowns. To minimize such flaring activity, depressurization manifold is provided to reuse the vented gas and thereby reducing the greenhouse emissions.

Efficient Resource Utilization

Standardized designs allow for optimized material usage during fabrication and construction. Modular, repeatable manifolds minimize waste, reduce the carbon footprint of logistics, and enhance the circular economy approach by simplifying dismantling and repurposing at end-of-life stages.

Lifecycle Sustainability

Standardized manifolds facilitate predictive maintenance and condition-based monitoring, which extends equipment lifespan and reduces the need for premature replacement. Fewer component failures also translate to reduced offshore mobilization, fewer helicopter trips, and less energy consumption across the asset lifecycle.

Ultimately, the environmental benefits of standardizing the gas lift manifold go hand-in-hand with operational gains—proving that sustainability and performance are not mutually exclusive, but in fact, mutually reinforcing.

Final Reflections and Strategic Implications

This initiative to standardize the gas lift manifold system represents a transformative approach to offshore infrastructure design. It is not merely about optimizing layout or minimizing pressure drop; it is about fundamentally rethinking how critical production systems are configured, operated, and maintained across their lifecycle.

- Long-term operational sustainability and a forward-thinking solution for optimized layouts.
- Benefit from lower operational costs, improved production reliability, and enhanced worker safety.
- Benefit from decarbonization by using depressurization manifold, cost control, and workforce transition.
- Prepares offshore operations for future expansion, digitalization, and automation.
- Elimination of liquid accumulation due to revised slope design.
- Reduction of time spent by Operators in high-risk zones.
- Training time reduction in onboarding for new operators.

This should be seen as a model of continuous improvement and engineering leadership in the offshore sector—setting a benchmark for how standardization, when implemented with vision and precision, can unlock significant value across the entire energy value chain.

Conclusion

This initiative demonstrates how technical excellence and business value are inherently connected. The optimized layout design reduces costs, increases safety, improves uptime, and enables future-ready capabilities. It transforms a routine component into a strategic asset—capable of delivering quantifiable impact on KPIs such as Total Recordable Incident Rate (TRIR), SAFETY, OPEX per barrel, Mean Time Between Failures (MTBF), and ESG performance.

From a strategic planning perspective, such a standardized system reduces the complexity of field development planning, accelerates project schedules, and supports more confident investment decisions due to reduction in CAPEX cost. It also builds a knowledge base that can be used across future projects, including expansion into new geographical markets or partnership-based joint ventures.

In summary, the standardization of the layout is a keystone in COMPANY's ongoing pursuit of excellence. It addresses present-day operational gaps, aligns with future technologies, and embodies the principles of sustainable, high-performance engineering. This is not merely a technical achievement, it is a strategic differentiator in a highly competitive and rapidly evolving global energy landscape.

Regulatory and Industry References

The design follows best practices and international standards, including:

1. ASME B31.3 – for process piping design, ensuring safe pressure containment and material selection.
2. API RP 14E – covering recommended practices for design and installation of offshore production platform piping systems.
3. HSE guidelines – for worker safety, access, and hazard control.
4. ISO 13703 – international standard for offshore production piping design, focusing on integrity, layout, and maintainability.
5. IEC guidelines – where applicable, to ensure integration with automation, electrical, and control systems.
6. Conceptual Design Study – Internal Document.
7. Findings from the Technical Alerts raised by Operations – Internal Document

By adhering to these guidelines, the standardized design ensures compliance with regulatory requirements and reinforces COMPANY's commitment to operational safety.

Moreover, standardization simplifies audit and inspection processes, as the consistent design facilitates quicker verification of compliance. This in turn reduces the administrative burden and enhances transparency across regulatory bodies.